THE NEW IBA SUPERCONDUCTING SYNCHROCYCLOTRON (S2C2): FROM MODELING TO REALITY

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Abstract

IBA is currently developing a compact superconducting synchrocyclotron (S2C2) for its new, low cost, proton therapy treatment system called ProteusOne®. The S2C2 is the first synchrocyclotron as well as the first superconducting accelerator ever produced at IBA. ProteusOne® is designed to be an alternative proton source for therapy, fulfilling the constraints of being a compact, reliable and operator free system, having low recurrent cost, full pencil beam scanning capabilities and short on-site deployment time.

Contrary to other IBA accelerators, the S2C2 is a non-isochronous machine, with different beam physics. As all IBA computational tools with regards to beam dynamics have so far been tailored to the principles of isochronous machines they had to be largely modified to deal with the new concept of the synchrocyclotron. Both modified and new calculation tools have been developed and include, for example, the ability to handle the longitudinal beam physics, the time-dependent RF frequency and non-linearities.

A brief overview of the accelerator design and its current status will be presented.

INTRODUCTION

The ProteusOne® is a single treatment room solution aimed at reducing the cost of protontherapy. In its superconducting form, it consists of the S2C2 superconducting proton accelerator, the new IBA compact gantry and a state-of the-art patient treatment room facility. The complete solution is designed for lower cost and compactness (the S2C2 diameter alone is a factor of ~2 smaller than the IBA Cyclone230) in order to help make proton therapy more widely accessible. The IBA superconducting synchrocyclotron for proton therapy has been designed in collaboration with AIMA Developpement [1], and the superconducting coil and cryostat have been designed and built by ASG [2].

The major advantage of proton treatment over conventional radiation is that the energy distribution of protons can be delivered in a precisely controlled range, known as the Bragg Peak, thereby reducing adverse effects to surrounding healthy tissues.

Conventional x-ray radiation treatment, given in sufficient doses, can also control many cancers. However, because of the inherent difficultly to adequately conform the irradiation pattern to the tumor, healthy tissues often receive a significant dose and can be damaged. Consequently, a less than desired dose is often used to reduce damage to healthy tissues and avoid side effects.

As a result of the dose-distribution characteristics of protons, a radiation oncologist can increase the dose to the tumor while reducing the dose to surrounding normal tissues. This allows the dose to be increased beyond that which less-conformal radiation will allow. The overall benefits are significantly reduced damage to healthy tissues and vital organs, fewer harmful side effects (e.g. secondary cancers or child growth abnormality), more direct impact on the tumor, and increased tumor control.

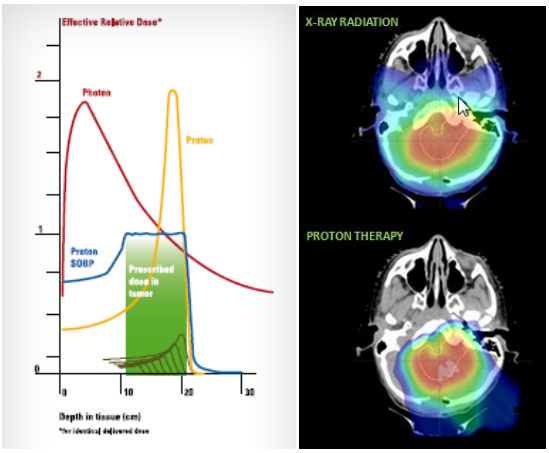


Figure 1: Left image: Proton beams release the majority of their destructive energy within a small range called the "Bragg Peak". This means that most of the energy can be deposited inside a tumor volume with little to no entrance and exit dose. This peculiarity of proton beams enables physicians to treat tumors with higher precision, safety and efficiency. Right image: Pediatric Ependymoma - the much improved dose comformality of proton therapy can be observed, leading to less damage to sensitive organs and healthy tissues.

S2C2 DESIGN OVERVIEW

Synchrotron versus cyclotron

Today comparatively few patients have benefited from proton therapy due to the few proton therapy treatment centers available. The significant financial investment and facility size required for such installations has been a limiting factor in the market so far. In order to make proton therapy more accessible to a larger public we need to reduce the cost and size of the accelerator. This in turn can be achieved by increasing the nominal magnetic field of the machine which can be done most effectively by

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using a superconducting coil. Although an isochronous cyclotron can be superconducting, for protons, focusing and extraction considerations limit the average magnetic field to about 2.5 Tesla for an isochronous machine. This therefore determines our choice of a superconducting synchrocyclotron for proton therapy: a machine which is compatible with high magnetic fields of 9 Tesla and above.

Important differences exist between the usual IBA isochronous cyclotron and a synchrocyclotron. For a synchrocyclotron, the magnetic field strength no longer increases with radius to compensate for relativistic increase in mass with speed, but rather it is rotationally symmetric and slightly decreasing with radius. The RF frequency becomes periodically modulated as opposed to constant, and the machine produces a pulsed beam instead of being continuous wave. Where little phase motion exists in the isochronous cyclotron (a few 10th's of a degree), longitudinal dynamics becomes a major driver for the beam physics in the synchrocyclotron with energy-phase oscillations bound by a separatrix. Beam capture now occurs during only a limited time window. Regenerative extraction must be used to recover the beam which has a very small turn separation at extraction. The extracted beam has a relatively low intensity typically on the order of a few hundred nanoamps, however, this is more than adequate for proton therapy applications. As well as the large external size reduction possible (~ factor of 2) for this machine, the central region is also greatly reduced by a factor of 5-10 compared to an equivalent isochronous cyclotron.

As all IBA computational tools had so far been developed for the principles of isochronous machines, such important differences have lead to a significant effort in modifying and developing new simulations tools to describe the behavior of the superconducting synchrocyclotron.

Key Parameters

Some general parameters of the IBA S2C2 are listed in Table 1.

KEY SUBSYSTEMS

Magnetic Design

The magnetic circuit has formed a central role in the design work for the S2C2 as many system parameters are involved in or impacted by the magnetic design. To name just a few of the considerations necessary for the magnetic design we can mention the forces (on the pole, coil, extraction system, return yoke and tie-rods), the design of the extraction system and harmonic coils, the median plane (vertical) error, stored energy, ramp-up time, eddy currents and AC-losses.

For a rotationally symmetric case the magnetic pole profile and the return yoke can initially be optimized in Opera2D [3]. However, to simulate the important effects of asymmetries in the system (such as the extraction system, cyclotron feet, channels in the yoke for extraction canal, vacuum pumping system etc) a detailed model in Opera3D [3] is required.

Table 1: Key Parameters of the S2C2

Parameter	**Value**
Maximum Energy:	230 MeV
Extracted Beam Current:	~ 200 nA
Size:	
- yoke radius	1.25 m
- pole radius	50 cm
- weight	50 tons
Coil:	NbTi
- ramp up rate / time	2-3 A per min / 4 hours
- stored energy	12 MJ
Magnetic field:	
- central	5.7 T
- extraction	5 T
Cooling:	Conductive (4x1.5 W cryocoolers)
- efficiency	~ factor 4000 (6.5kW per refrigerator)
- initial cooldown	12-14 days
- cooldown after quench	less than 2 days
Pulse:	
- repetition rate	1000 Hz
- pulse length	~ 7 microsec
RF system:	1-Dee system
- frequency	92 - 62 MHz
- voltage	10 kV
- power	few kWatt
- RF-bias	~ 1 kV
Extraction:	Passive regenerative
- efficiency	50%
Ion source:	PIG Cold Cathode
Central region:	Removable module
- 1st orbit radius	3 mm
- source diameter	< 5mm

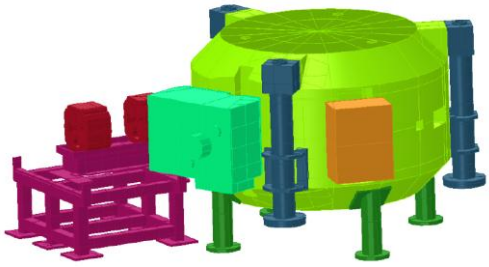


Figure 2: Elements included in the Opera3D[3] magnetic model of the S2C2: included are the Yoke, Poles, Coils, Yoke penetrations, Extraction system (regenerator, channels, first harmonic correctors), Harmonic coils, External systems (Cyclotron feet, Yoke lifting system, Shields (cryo-coolers + rotco) and External quadrupoles).

Due to saturation of the yoke iron we also found that external systems had to be included in the 3D magnetic design studies, and also that it was necessary to shield the cryo-coolers and rotating capacitor against eddy current heating.

A new mapping system has been developed at IBA specifically for field measurements of the S2C2 machine. One of the major differences with existing mapping systems at IBA is the replacement of the Hall probe by a search coil. The Hall probe in existing systems is calibrated in a field region from 0.35 to 2.8 T, which suffices to reach a relative precision of 1E-5 in the mapping of the existing normal conducting isochronous cyclotrons at IBA. In the S2C2, the field values range from 4.6 up to 5.7 T and large field gradients exist in the extraction region. These gradients need to be measured with high precision to ensure proper extraction from the machine. For a fast, stable and accurate mapping of the S2C2, a "flying" search coil, connected to an integrator has been designed and the results compare very well with the magnetic field simulations performed in Opera3D.

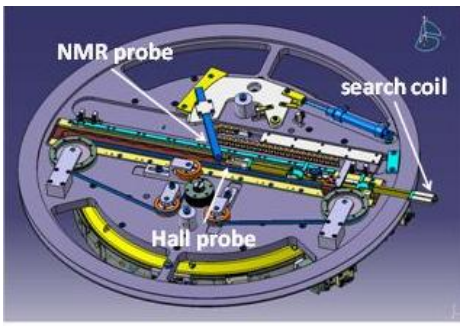


Figure 3: The mapping wheel with "flying" search coil developed for the precise measurement of the S2C2 magnetic field [6].

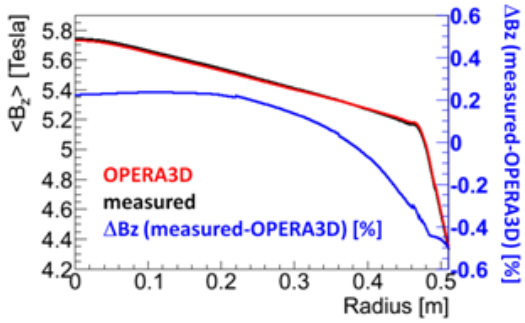


Figure 4: The magnetic field profile in the S2C2 as a function of machine radius. The black curve shows the field result measured with the search coil, while the red curve shows the field calculated in Opera3D. The blue curve shows that the results correspond very well, with an observed difference of less than 0.5%.

Extraction System

Acceleration in a synchrocyclotron occurs slowly and with a much smaller turn separation than in an isochronous cyclotron. For this reason, it is not possible to use electrostatic deflection as an extraction method - we must use resonance extraction based on the 2Qh = 2 resonance. The S2C2 resonant extraction system is a fully passive system using only soft iron (no power requirement). A strong local field bump produced by the regenerator increases the horizontal betatron frequency and locks it to unity, creating an unstable orbit. The resonance shifts the radius of the unstable orbit exponentially towards the extraction channel. The gradient corrector and permanent magnet quadrupole (PMQ) horizontally focus the beam when passing through the fringe field. Five corrector bars are located near the regenerator and extraction channel to reduce the strong first harmonic error which would leave an undesired long tail on the field bumps. Care must also be taken to avoid the Walkinshaw resonance at Qh = 2Qv which would result in beam loss via vertical blow up. The extraction in the S2C2 occurs in approximately 20 turns.

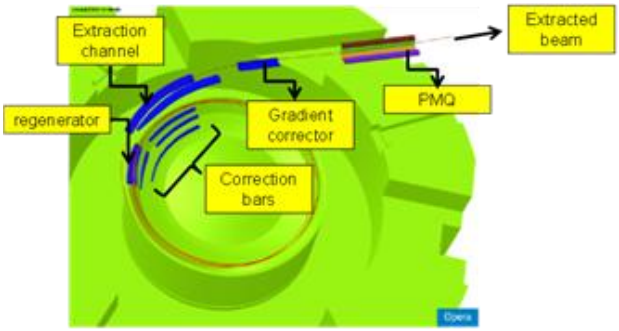


Figure 5: The Opera3D model of the S2C2 resonant extraction system.

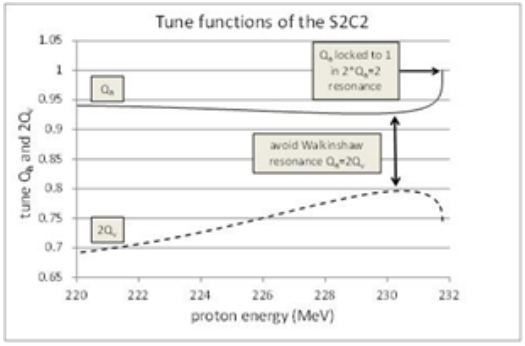


Figure 6: The horizontal and vertical tune functions for the S2C2 as a function of energy. At the extraction radius Qh becomes locked to unity. The Walkinshaw resonance is avoided.

RF System

As for all synchrocyclotrons, the RF frequency must vary during the acceleration: in our design the frequency spans from 92 MHz to 62 MHz. This is realized thanks to an innovative and patented rotating capacitor (Rotco) design. The RF power is supplied by a triode-based oscillator capacitively coupled to the resonator. The RF

resonator as well as the oscillator have been initially optimized using a simplified model in Excel and then refined using CST Microwave Studio® [4] [5].

The synchrocyclotron RF system accelerates the particles on the first harmonic mode. It gives an RF frequency of 87.5 MHz in the machine centre and 63.3 MHz at the extraction radius. The principle of the RF resonator is based on a half-wave transmission line with a 180° Dee electrode attached at one end and a rotating capacitor connected at the opposite end. The rotation occurs at 7500 rpm with a repetition rate of 1 kHz. The Dee and its resonator are DC biased at ~1 kV in order to avoid multipactoring. Resonator side stubs are used to slightly tune the maximum frequency of oscillation and hence the rate of frequency change during capture in the machine centre. As some parts of the Rotco rotate at high speed, it is located outside of the cyclotron and is further shielded from magnetic fields to avoid eddy current heating.

At this stage all the mechanical parts for the RF system have been realized, installed in the S2C2 and tested at full voltage. Fine tuning of the Rotco stator blades is in progress in order to obtain a df/dt profile that keeps the size of the separatrix constant for a given Dee voltage.



Figure 7: The S2C2 RF system on the test-bench in the IBA assembly hall.

Ion Source

For fast and precise pencil beam scanning treatment (PBS) we require a high dynamic range in the charge content (picoCoulombs) of each pulse. A factor of about 100 in the charge content, is necessary in order to deliver the range of treatment doses clinically required. In addition, we must ensure good pulse repeatability with an error of less than 1%, fast source rise times of less than 10 microseconds and sufficient DC output current (typically ~ 1 mA).

For the S2C2 we have chosen to use a cold cathode PIG ion source as this has the advantages of faster pulse moderation, longer source lifetime and increased pulse stability due to its better temperature independence. In order to maintain the low temperature of the cathode we opt for a small pulse duration of about 50 microseconds. The pulse frequency is 1kHz and is synchronised with the rotating capacitor.

The source has been studied and optimised by AIMA Developpement [1] in a 1 Tesla test stand which demonstrated an extracted DC H+ current of up to 6 mA (a factor 6 more than required). Testing at 1 Tesla also showed the importance of wiring and cathode capacity for current optimisation and that pulse stability was less than optimal for low arc current tests. On the other hand, results subsequently repeated at IBA in the S2C2 magnet at 5 Tesla showed very good arc-current stability and good reproducibility of the plasma impedance measurement performed by AIMA indicating that the source works well at the higher fields required by the accelerator. Although extracted beam current tests are expected shortly, we believe that arc current tests at 5 T are very likely to be representative of the extracted beam results from the central region.

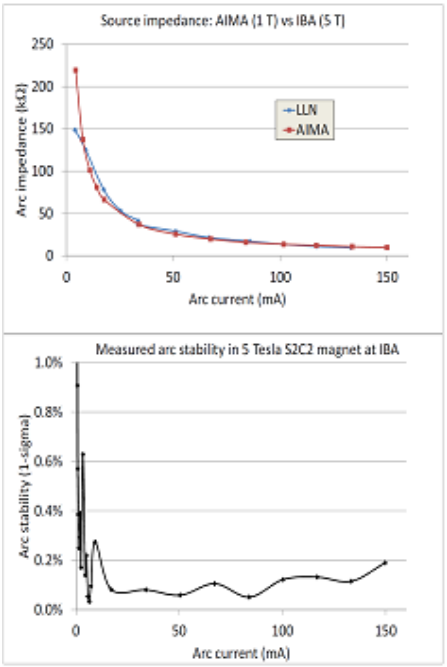


Figure 8: Top graph: Test measurements of the plasma impedance at AIMA Development (1 Tesla) and IBA (LLN; 5 Tesla) show good correlation indicating that the source works well at the higher fields required by the S2C2 accelerator. Bottom graph: The measured arc stability is representative of beam pulse reproducibility from the S2C2. The test results above are performed in the S2C2 magnet at 5 Tesla in the IBA assembly hall. The results show a high level of arc-current stability.

Central Region

The central region of the S2C2 is extremely compact with a chimney diameter smaller than 5 mm and a first orbit radius of about 3 mm. The vertical gap in the centre of the machine is about 6mm and tapers in a cone shape to enable better vertical focusing and transit time in the first turns of the acceleration. The first 100 turns in the central region extend to a radius of about 3 cm.

The central region and ion source are removable as one sub-system for easy maintenance and precise re-alignment. A bias voltage of ~1kV is applied to the dee and counter-dee to suppress multipactoring – an electron multiplication phenomenon observed in RF cavities which can prevent achievement of the required voltage and field if not suppressed.

The central region has been accurately simulated in Opera3D [3] and in-house orbit tracking calculation tools. Such calculations have been used to optimize beam acceptance and transmission and to ensure good beam centering and quality. Particle tracking calculations have demonstrated a small radial emittance in the centre of the S2C2 indicative of good horizontal beam quality.

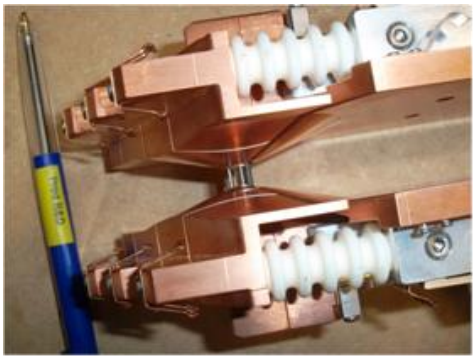


Figure 9: The S2C2 central region and ion source are removable as one sub-system for easy maintenance and precise alignment.

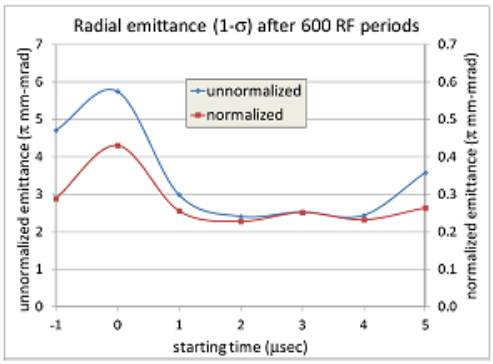


Figure 10: Particle tracking calculations have demonstrated a small radial emittance in the centre of the S2C2 indicative of good horizontal beam quality.

CURRENT STATUS AND PERSPECTIVES

In recent months the magnet has been successfully cooled down to 4.2K and ramped up to the central design field of 5.7 Tesla. The coil has been well centered in the cryostat and vertical forces on the cold mass have been strongly reduced by the use of external shims. The magnet has been fully mapped and found to correspond very well with calculated field maps. No additional shimming is necessary to correct the magnet field profile.

The ion source, central region, vacuum system, rotco and RF-system have all been tested and are now installed in the machine. The S2C2 is fully assembled and installed in the IBA shielded bunker for the start of beam testing, with first beam anticipated before the end of August 2013.

CONCLUSIONS

Since the 2008 preliminary studies, all subsystems needed for the IBA S2C2 have been developed, keeping in mind their integration in the whole system. This approach, combined with good team communication and careful design reviews, detected any issues and optimized the machine for good treatment quality. The choice of a new machine to IBA - a superconducting synchrocyclotron – as well as our most compact proton therapy machine, required the development of a new suite of calculation tools and consideration of many new issues, such as the superconducting coil, very small central region, regenerative extraction and increased magnetic forces, to name but a few.

Accurate field mapping has been performed at IBA and results are in very good agreement with the model predictions. The beam testing of the prototype S2C2 is scheduled for summer 2013 and external commissioning is expected by 2014.



Figure 11: The S2C2 in the IBA assembly hall, July 2013.

REFERENCES

[1] AIMA Developpement, Nice, France.

[2] ASG Superconductors, S.p.A., Genova, Italy.

[3] Cobham Antenna Systems, Vector Fields Simulation Software, Kidlington, UK.

[4] CST Microwave Studio® 2011-SP6, CST Computer Simulation Technology AG.

[5] Poster presentation: "Design of the New IBA S2C2 RF System", M. Abs, Y. Jongen, J. Mandrillon, P. Mandrillon, P. Verbruggen, J-C Amélia, presented at the European Cyclotron Progress Meeting, 2012.

[6] Poster presentation: "Developments on the S2C2 mapping system", J. Van de Walle, C. L'Abbate, Y. Paradis, presented at the European Cyclotron Progress Meeting, 2012.